

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

Duration: 30 Theory hrs + 40 Lab hrs + 20 Self-Learning hrs (Total: 90 hrs)

Objective: To introduce the students to Network Defense and Countermeasures.
This includes the following:

- Network Security Concepts,
- Firewalls,
- IDS & IPS, and
- VPN

Prerequisites: OS and Network Concepts

Evaluation method:

Theory Exam: 40% weightage.

Lab Exam: 40%weightage.

Internal Assessment: 20%weightage.

List of books / Other training Material:

Courseware: Cryptography & Network Security: Principles and Practices by William Stallings.(Pearson Education), 7th Edition

Reference:

- Fundamentals of Network and Security: Eric Maiwald/TMH

Session 1: (2T + 3SL)

Theory:

- Introduction to Information Security
- Why Information Security?
- Security: The money factor involved
- Internet Statistics - Study from a security perspective
- Vulnerability, Threat and Risk
- Qos

Self-Learning:

- Risk identification for small website
- Explore OWASP Top 10 (latest version) from a risk perspective
- Look into NIST Cybersecurity Framework
- Case Study: "Top 3 breaches in last 5 years and their root cause"
Analyze:
 - ◆ Attack Vector
 - ◆ Vulnerability exploited

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

- ◆ Financial & reputational impact

Session 2: (2T+4L)

Theory:

- Risk Management, Exposure and Countermeasure
- Firewall
- De-militarized Zone
- Two methods of implementing firewall

Lab 1: (4L)

- Install and configure pfSense as a firewall on your network.
- Set up a DMZ using a virtual environment to host a web server with pfSense.
- Configure and compare packet-filtering and proxy firewalls using iptables and Squid.

Session 3: (2T + 4L)

Theory:

- Packet Filtering
- Screened Host Firewall
- Stateful Inspection Firewall
- NextGen Firewall app controls
- iptables - Linux Firewall

Lab 2: (2L)

- iptables - Rule Processing
- Default Policy
- iptables - Predefined tables
- INPUT
- OUTPUT
- FORWARD

Lab 3: (2L)

- Iptables / netfilter/ Xtables-Addons
- verifying iptables / netfilter
- internal working
- iptables / netfilter - concept of targets
- DROP
- ACCEPT
- iptables - SPI Firewall
- Setting up a SPI Firewall - Standard installation

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

- Iptables IPv6 Rule Management

Session 4: (4L + 4 SL)

Lab 4: (2L)

- Advanced iptables
- Loading of modules
- Geo IP Blocking using Xtables-Addons

Lab 5: (2L)

- Access control using iptables
- Internet sharing Using Iptables

Self-Learning: (4SL) (SL is based on session 2- 4 & lab 1- 5)

- Make comparison between Stateless vs Stateful firewalls
- Make difference between traditional firewall vs NGFW
- Study real firewall rule misconfiguration incidents
- Write scripts to automate iptables
- Case Study: How a misconfigured firewall led to data exposure
- List out the GeoIP blocking use cases

Session 5: (2T+2L+2SL)

Theory:

- Wireshark
- Create a filters for data collection and display
- Examine real-world packet captures

Lab 6: (2L)

- Wireshark
- Examine real-world packet captures
- Iptables Port Forwarding
- Iptables use case: Fail2ban

Self-Learning: (2SL)

- Study common attack patterns in packet captures
 - ◆ Study TCP packets(TLS, HTTP, Kerberos) using Wireshark.
 - ◆ Define some rules in IP tables to block IP's of particular countries
- Capture network traffic and
 - ◆ Apply display & capture filters
 - ◆ Identify at least 3 suspicious patterns
- Analyze QoS parameters (Latency, Jitter, Packet loss)

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

Session 6: (2T+2L+2SL)**Theory:**

- Linux Software Firewall(ClearOS Pfsense)
- Nginx & Squid Reverse Proxy
- UTM
- Server Load Balancing

Lab 7: (2L)

- Nginx & Squid Reverse Proxy
- Configure reverse proxy URL using regex
- Server Farming

Self-Learning: (2 SL)

- Study UTM vs Next-Gen Firewall
- Explore Reverse Proxy security use cases
 - ◆ SSL termination
 - ◆ Web Application protection
- Write a use case comparison between Load balancer vs Reverse Proxy
- Case Study: How reverse proxies protect against
 - ◆ DDoS
 - ◆ Application layer attacks

Session 7 & 8: (4T + 8L)**Theory:**

- VPN – Introduction
- VPN protocols/characteristics
- VPN Functions
- Types of VPN
- SecureVPN
- Trusted VPN

Lab 8: (2L)

- OpenVPN configuration in both Linux & Windows
- Site to Site Connectivity

Lab 9: (2L)

- Pfsense OS - Installation and Configuration
- Installation of Pfsense OS and Basic
- Configuration

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

Lab 10: (2L)

- UTM Configuration
- Basic Setup of UTM

Lab 11: (2L)

- Configuration for access control and Firewall features (UTM)
- UTM VPN Configuration

Session 9: (2T + 2L + 3SL)**Theory:**

- Hybrid VPN
- IPsec
- Tunnel mode/transport mode
- Ipv6 VPN
- Split Tunnel full tunnel VPN

Lab 12: (2L)

- PPTP/SSTP VPN Setup
- VPN Configuration under Windows 2016 using RRAS

Self-Learning: (3 SL) (SL based on session 7-9 & lab 8-12)

- Configure L2TP VPN
- Compare Full Tunnel vs Split Tunnel
- Study VPN vulnerabilities & attacks
- VPN exploitation incidents
 - ◆ Misconfigured VPNs
 - ◆ Credential reuse attacks
- Configure Pfsense for routing data from one machine to another (consider machines on 192.168.2.1/24 and 192.168.1.1/24 subnets).

Session 10: (2T)**Theory:**

- Introduction to IDS and IPS
- IDS / IPS
- Types of Attacks
- Security Events
- Vulnerability/design/implementation
- Tcpdump and its basic usage

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

Session 11 & 12: (4T + 8L +3SL)

Theory:

- Attacks-traditional/distributed
- Intruder types
- Types of IDS
- IPS categories
- Defence in depth
- IDS and IPS analysis scheme
- Detection methodologies
- Principles of IDS
- Threat Hunting model

Lab 13: (4L)

- Install and compare Snort (NIDS) and OSSEC (HIDS) on your network.
- Set up Suricata as an IPS and configure rules for network intrusion prevention.

Lab 14: (4L)

- Analyze network traffic logs from Snort and Suricata to identify and respond to threats.
- Perform threat hunting using the ELK stack (Elasticsearch, Logstash, Kibana) to detect anomalies.

Self-Learning: (3 SL) (SL based on session 10-12 & lab 13-14)

- Tcpdump installation, verify its installation and capture traffic on an interface
- Study Signature-based vs Anomaly-based detection
- Explore MITRE ATT&CK Framework
- Study about Distributed Honeynet system developed by C-DAC
- Look for UAC (URL Analyzer and Classifier)
- Case Study: How IDS/IPS failed or succeeded in detecting attacks
- Write 5 basic snort rule for
 - ◆ Port scan
 - ◆ ICMP flood
 - ◆ Suspicious HTTP activity

Session 13 & 14: (4T + 2L + 2SL)

Theory:

- Symptoms of attacks
- Tired architecture
- Sensors-network/host based
- Denial of services

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

- Dos & DDos Mitigation
- Sensor Deployment
- Agents
- Functions of IDS agents
- IDS Manager

Lab 15: (2L)

- Identify and document symptoms of various attacks using Wireshark..
- Set up and configure network-based sensors with Snort and host-based sensors with OSSEC

Self-Learning: (2 SL) (SL is based on session 13-14 & lab 15)

- Design and implement a three-tier architecture in a virtual environment
- Study modern DDoS attack trends
- Explore Cloud- based DDoS mitigation
- Implement rate limiting techniques using iptables to mitigate DoS/DDoS attacks
- Design a DDoS mitigation strategy
 - ◆ Network layer
 - ◆ Application layer
- Explain defense-in-depth approach

Session 15: (2T + 2L)

Theory:

- Introduction of Log Analyser
- Log
- SIEM Log Correlation and event triggering
- Introduction of SIEM
- SIEM Log Forwarding Configuration
- SIEM Log Correlation and event triggering

Lab 16: (2L)

- Snort
- Writing Basic Snort Rules
- Syslog Server
- BASE
- Configuration and deployment of SIEM

Suggested Teaching Guideline for
Network Defense & Countermeasures
PGCP- ITISS February 2026

Session 16: (2T + 2L + 1SL)

Theory:

- Testing Snort in both Windows & Linux
- IDS architecture
- Bypassing an IDS

Lab 17: (2L)

- Nagios
- Nagios Sensor Configuration (Windows & Linux)
- Implement Nagios up/Down of RAM

Self-Learning: (1 SL) (SL based on session 15-16 & lab 16-17)

- Implement Nagios with Email services
- Implement Nagios up/Down of Hard Disk
- Study SIEM use cases
- Explore SOC workflows
- Create a log correlation scenario
 - ◆ Firewall log
 - ◆ IDS alert
 - ◆ System log
- Explain how SIEM detects an attack chain